

Machine learning-based prediction model of wind turbine power generation

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ABSTRACT

Renewable energy sources have become central to the transition toward cleaner energy systems, with wind energy demonstrating the most rapid global growth since 1990. However, its production is inherently dependent on variable and uncontrollable factors such as weather conditions and wind dynamics. In this work, we analyze a dataset spanning two and a half years, collected from wind turbines, and apply extensive exploratory data analysis and preprocessing to enable accurate forecasting of wind power generation. Initially, the dataset was evaluated using multiple regression models for baseline predictions, while the Prophet model was employed to extract long-term trends and seasonality. The processed data were then integrated and used as input for CatBoost and Random Forest models, incorporating a windowing mechanism informed by autocorrelation and partial autocorrelation analysis to optimize temporal dependencies. Forecasting was conducted across three horizons: 15 min, 1 day, and 1 week ahead. The proposed hybrid approach achieved a root mean square error of 30.6 for 15 min forecasting, 50 for one-day forecasting, and 41 for one-week forecasting, representing at least a 50% improvement over the best standalone regression. Results further confirm the expected trend that longer forecasting horizons increase RMSE and reduce R^2 , due to resampling constraints and the need for more extensive input data. Nonetheless, the hybrid methodology consistently outperformed standalone models, demonstrating stability and robustness across different horizons. By leveraging the complementary strengths of multiple regressors within a unified framework, this study highlights the potential of hybrid machine learning approaches to significantly enhance the predictive accuracy of wind energy forecasting.

1. Introduction

Our planet is facing pressing and unprecedented challenges, ranging from weather events and global warming fueled by carbon emissions to an increasingly complex energy market and soaring energy prices. These combined risks have escalated dramatically in the last few decades, driven by population growth and the increasing energy demand met largely by fossil fuels. In 2020, fossil fuels still dominated global energy consumption at 84.3% (Holecek et al., 2022).

Recent analyses indicate that adhering to the Paris Agreement's goal of limiting global temperature rise to 1.5 °C necessitates tripling the installed capacity of renewable energy by 2030. This step is critical in mitigating the severe impacts of climate change (Huang and Zhai, 2021). Renewable energy systems have emerged as a promising solution to environmental concerns, particularly those related to energy. These systems attracted researchers for three main reasons: a clean energy source with minimal emissions compared to fossil fuels, an

infinite prospect for utilizing natural resources, and the possibility of integration in every country, including developing ones, because these sources are available at almost every point on the planet with some degree of variation. Wind energy, solar radiation, and water are the primary sources of these systems, with others such as biomass and geothermal being increasingly used (Allal et al., 2024). Wind energy has recently seen widespread use worldwide, with its enormous potential prompting further development of harnessing technology.

Wind energy is now more affordable than fossil fuels. According to Lu and McElroy (2017), in most regions of the world, offshore wind energy has become cost-competitive with fossil-fuel-based electricity, and onshore wind energy is expected to follow suit in the next decade. This trend explains why developed countries are eager to expand their wind power installations. As of 2023, data from the International Renewable Energy Agency (IRENA) shows that China has the largest installed wind energy capacity with 365 GW, followed by the United

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Table 1
List of acronyms.

Acronym/Abbreviation	Full form
ML	Machine Learning
NREL	National Renewable Energy Laboratory
KNN	K-Nearest Neighbors
SVM	Support Vector Machine
SVR	Support Vector Regression
PSO	Particle Swarm Optimization
NN	Neural Network
LSTM-EFG	Long-Short-Term Memory - Enhanced Forget Gate
SCADA	Supervisory Control and Data Acquisition
CNN	Convolutional Neural Network
DL	Deep Learning
GRU	Gated Recurrent Unit
DCNN	Deep Convolutional Neural Network
WPT	Wavelet Packet Transformation
DSM	Demand Side Management
MAPE	Mean Absolute Percentage Error
WTG	Wind Turbine Generator
RBFNN	Radial Basis Function Neural Network
KPSS	Kwiatkowski–Phillips–Schmidt–Shin
ADF	Augmented Dickey–Fuller
DT	Decision Tree
RF	Random Forest
XGB	Extreme Gradient Boosting
LGBM	Light Gradient Boosting
MLP	Multi Layered Perceptron
ICE	Individual Conditional Expectation

States with 140 GW, Germany with 66 GW, India with 41 GW, and Spain with 29 GW. The exponential growth in wind energy is largely attributed to extensive research and development efforts. Technological advancements and the evolution of supply chains have driven down the cost of wind turbine components during manufacturing. However, the primary challenge now lies in accurately forecasting wind energy.

Machine learning (ML) can be instrumental in meteorological forecasting, especially in predicting wind power. By analyzing diverse data, such as wind speed, temperature, and humidity, these algorithms enhance accuracy, adapt to changing weather patterns, and improve future power generation estimates (Colston et al., 2018). In wind power forecasting, ML models factor in variables such as wind speed, direction, topography, and atmospheric pressure, using historical data to predict future production. Furthermore, ML enables ensemble forecasting models by combining predictions for heightened accuracy, crucial in meteorology's unpredictable nature (Xie et al., 2018; Zhang et al., 2019; Aggarwal and Chaturvedi, 2021).

In the literature, both traditional ML models and deep learning (DL) techniques have been employed to forecast wind power, typically focusing on two primary tasks: regression analysis and classification. Generally, ML models have demonstrated the ability to effectively capture complex dependencies and relationships among multiple features, leading to strong prediction performance. In Treiber et al. (2016), short-term wind power forecasting was addressed using NREL data for a wind park and a single turbine in a regression challenge. Multiple regressors were used, including KNN, SVM, and linear regression; in a 30 min horizon forecasting, the SVR algorithm performed best, with an accuracy of 24%. In Zameer et al. (2015), a new hybrid methodology called ML-STWP was proposed, which uses ML techniques for feature selection and regression, as well as SVR as an auxiliary prediction algorithm and PSO with NN as final predictors. Irrelevance and redundancy filters were employed to select appropriate atmospheric properties. The results demonstrated the suggested strategy's efficacy compared to prior ones on a daily forecasting horizon. In Yu et al. (2019), a better long-short-term memory (forget gate LSTM-EFG) was presented to forecast wind power using data pre-processing and feature generation approaches. The model was assisted by wind data clustering to improve prediction performance. The results showed that the suggested model could predict wind power at specific times. In Leahy et al. (2016),

SCADA system data were used to diagnose and analyze problems and alarms for a southeast island wind turbine. The proposed system used ML binary classifiers, specifically SVM, to classify fault and fault-free samples before deploying them for predictive diagnosis. The results demonstrated the system's power in detecting a wide range of defects. In Nikolaidis (2022), wind speed and direction, ambient temperature, relative humidity, renewable capacity, and renewable energy sources were combined in multiple scenarios to realize a decision tree for a continuous response. The aim was to predict the generated power in distribution networks for five wind farms in different regions. The results demonstrate the significance of these characteristics in reducing prediction errors and the power of the predictive model. In Wang et al. (2017), probabilistic wind power forecasting was the focus. A novel DL ensemble-based method that combines a CNN learning algorithm with a wavelet transformation for frequency data decomposition was used to achieve this. The method produced a competitive performance compared to earlier techniques that employed shallow networks and SVMs. In Lin and Liu (2020), wind power forecasting was performed using a very high-frequency SCADA database. A neural network was built in a forecasting system made up of a Pearson feature correlation investigation, and various parameters were examined at various altitudes. The outcomes demonstrated that the suggested approach can maintain high accuracy with little computing expense, allowing it to be used for additional wind turbines or scaled up to wind farms. A hybrid DL NN for day-ahead wind power generation forecasting was presented in Hong and Rioflorida (2019). To extract uncertain wind power characteristics, a combination of a CNN for wind power characteristic extraction, RBFNN, and a double Gaussian function as activation functions for the RBFNN was used. A seasonal comparison of the data was done, and NN parameters were investigated for greater accuracy. The temporal convolutional neural network technique was used in Lin et al. (2021) to study long-term 24–72 ahead wind power predictions. The model was trained using Turkish SCADA wind power data. For the targeted horizon, a MAPE of 5.13% was attained. Moreover, the model performed better than other DLs because of its higher prediction accuracy and lower convergence error than LSTM and GRU. In Mujeeb et al. (2019), wind power was predicted using a DCNN and a wavelet packet transformation. While the DCNN was utilized for prediction, WPT was used to dissect previous wind signals, and afterward, the model results were used to obtain a day-ahead power demand and price scheme. Numerical results in the forecasting window demonstrated the effectiveness of the suggested DSM technique.

This work aims to create an ML model for wind power forecasting. Historical data from a wind power facility and weather data from neighboring meteorological stations are used to train the model. The model's ability to reliably anticipate wind power output for short-term horizons (day-week ahead) is assessed. Indeed, the task's specificity makes it complicated; nonetheless, this can be accomplished in line with objectives such as identifying influencing parameters on wind power output (e.g., temperature, wind speed). Second, a high-performance ML model capable of accurately predicting the target variable is built. The last point was only possible by testing a large number of regressors. Third, an evaluation process is conducted by comparing the results obtained with existing approaches, such as auto-regressive models or traditional numerical weather prediction models, as well as a comparison with previous work results. Finally, the sensitivity of the best models to input parameters is investigated, with hyperparameter tuning employed for improved accuracy. The study's findings have practical implications for wind farm operators, electricity market participants, and policymakers. Accurate wind power forecasting can help reduce the uncertainty and risk associated with wind power output and enhance wind power penetration in the electricity mix.

The remainder of this paper is organized as follows. Section 2 details the proposed framework, including data presentation, exploratory

Table 2
Dataset feature description.

Feature	Description	Physical unit
Active Power	The amount of electrical power produced by the wind turbine.	Kilowatts (kW)
Ambient Temperature	The air temperature around the wind turbine	Degrees Celsius (°C)
Bearing Shaft Temperature	The temperature of the shaft bearing inside the wind turbine.	Degrees Celsius (°C)
Blade1PitchAngle, Blade2PitchAngle, Blade3PitchAngle	The angle of pitch of each wind turbine blade.	Degrees (°)
Control Box Temperature	The temperature of the control box that manages the wind turbine.	Degrees Celsius (°C)
Gearbox Bearing Temperature	The temperature of the gearbox bearing inside the wind turbine.	Degrees Celsius (°C)
Gearbox Oil Temperature	The gearbox oil temperature inside the wind turbine.	Degrees Celsius (°C)
Generator RPM	The number of revolutions per minute of the wind turbine generator	Revolutions per minute (RPM)
Generator Winding1Temperature, GeneratorWinding2Temperature	The temperature of each winding of the wind turbine generator.	Degrees Celsius (°C)
Hub Temperature	The temperature of the hub that connects the blades to the wind turbine.	Degrees Celsius (°C)
Main Box Temperature	The temperature of the main control box that manages the wind turbine.	Degrees Celsius (°C)
Nacelle Position	The position of the nacelle, which is part of the wind turbine that contains the gearbox and generator.	Degrees (°)
Reactive Power	The amount of reactive power produced by the wind turbine.	Kilo-Volt-Ampere-Reactive (kVAR)
Rotor RPM	The number of revolutions per minute of the wind turbine rotor.	Revolutions per minute (RPM)
Turbine Status	The status of the wind turbine, such as running or stopped.	Categorical variable Run Stop
WTG	The wind turbine generator number.	Categorical variable WTG01
Wind Direction	The direction from which the wind is blowing.	Degrees (°)
Wind Speed	The speed of the wind	Meters per second (m/s)
Date: year, month, day, hour, and minute	The date and time of the data point.	Time units

techniques, models, evaluation metrics, and the hybrid model. Section 3 presents the results and discussion, covering data analysis findings, model performance, sensitivity analysis, and confidence in interpretable AI. Finally, Section 4 draws the conclusions of this work and highlights key insights and future perspectives. Table 1 gives the list of acronyms used in the paper.

2. Proposed framework

This section presents the dataset and the employed data manipulation techniques, analyzes the features and the employed ML regression models and metrics, and describes the proposed framework.

2.1. Data description

The data set as described in TheForceCoder (2023) is a two-and-a-half-year data set of wind turbines with a 10 min interval composed of numerous characteristics of the weather, the turbine, and the rotor between the timestamps of January 2018 and March 2020. The dataset features are described in Table 2. The dataset will be utilized initially to train multiple regressors, aiming to predict wind active power. Subsequently, the best regressor will be chosen based on specific metrics described in the following sections. The performance of the selected regressor will be compared to that of a hybrid system, which incorporates trending and seasonality extraction using the Prophet model. The analysis demonstrates that this approach significantly improves the accuracy of the prediction results.

2.2. Data wrangling and feature engineering

A crucial stage in getting data ready for analysis is data wrangling, commonly referred to as data cleaning. It guarantees that the data is correct, consistent, and analytically useful. Accurate analysis is made possible by the data's clear, orderly, and useful structure. Scrutinizing the data and spotting any mistakes, discrepancies, or missing values is the first stage in data wrangling. To do this, it may be necessary to look for outliers, recognize inaccurate or lacking data, and ensure it is properly structured. The next step is to clean the data if any problems are found. This can involve a variety of tests, such as eliminating duplicates, filling out missing values, and fixing errors. The data must then be transformed into a format that is easily analyzed. This may

entail adding new variables, averaging data, or fusing different datasets (Feature Engineering). To arrange the data for analysis, it may need to be sorted, important variables should be selected, and visualizations created to aid in understanding the data.

In this work, the data wrangling steps include the following:

- **Indexation:** Because the task at hand is to predict the power generated by a wind turbine, the data was provided as a time series, but it was indexed using integer indexation. After importing the data, the first step was to set the index to a date-time index so that every sample was handled according to its corresponding time, simplifying subsequent re-sampling operations. Resampling transforms data into time lags (daily, weekly, monthly, yearly, etc.). This is critical for plotting and data comprehension, particularly in studying phenomena related to the season (time in general).
- **Categorical Encoding:** The datasets studied include numerical and categorical features. The latter refers to variables that are expressed in a non-numerical format, such as the type of turbine or its number (first, second, third, etc.). These features must be encoded in a regression task, where the predicted value is numerical. For this purpose, the label encoder from the scikit-learn preprocessing module was used.
- **Filling Missingness:** It is a crucial step in data manipulation that seeks to replace missing values in a dataset with relevant values. Missing values in a dataset can occur for various reasons, such as human mistakes during data collection, corruption, or loss during transmission. Addressing missing values before performing any analysis is critical since they can degrade algorithm performance and lead to biased results. In this work, recognizing the dependencies between each current value and its lagged counterparts up to three time points, as indicated by the PACF analysis, missing data were imputed by averaging the values of the preceding three time points.
- **Data Information:** This is a simple visualization of the dataset's content, which includes knowledge of the min and max values in each series, the standard deviation, the quartile distribution, and a count of the values in each dataset. This last point is as important as the others because some cases, such as series with a lack of data or merging two datasets with different numbers of values, may generate errors in the subsequent steps.

Additional tests were carried out in the exploratory data analysis to better understand the series and, more importantly, to look for correlations between the characteristics and the target variable. These tests included:

- **Outlier Study:** Outliers are observations that deviate significantly from the typical range of values in a dataset. They can arise from measurement errors, unusual events, or data entry issues and may occur in both numerical and categorical variables. For instance, in a wind energy dataset where wind speed is measured in m/s, the daily measurements [3.2, 3.8, 4.0, ..., 7.5, 30.0] clearly indicate that the last value, 30 m/s, is an outlier, likely due to an anomalous measurement or extreme weather conditions. Such outliers can substantially bias model training and degrade forecasting performance. Therefore, their detection and treatment are critical during preprocessing. In this work, outliers were identified using a rolling-window z-score approach, which quantifies how many standard deviations each value deviates from the local mean. Values exceeding a threshold of 3.0 were replaced with the median of a ± 3 -hour neighborhood, ensuring robustness while preserving the temporal structure of the data.
- **Data Distribution:** It entails examining the statistical features of the data and how they are dispersed across various factors. Because of the importance of assessing frequency distribution, central tendency, and dispersion, the distribution of the data can significantly impact the choice of ML methods and the performance of the models. This step investigates trends, patterns, and the relationships between characteristics for feature engineering and model selection. Normal data is suited for models such as linear regression, whereas skewed data is more associated with other models such as SVR; nonetheless, there are approaches for changing data from one distribution to another using mathematical functions. This stage was tackled using histograms.
- **Correlation Study:** The process of assessing the link between variables in a dataset is known as correlation. It entails determining the magnitude and direction of a linear relationship between two or more variables. It is significant since it can aid in determining which traits are highly related to the target variable and, hence, most valuable for prediction. Features with a strong correlation to the target variable are more likely to be predictive than features with a weak correlation. The Pearson correlation coefficient is the most commonly used method for measuring correlation. This coefficient measures the degree and direction of the linear relationship between two continuous variables. However, two other correlation approaches, Spearman and Kendall, were tested. This allowed us to eliminate the less frequently linked features from the three techniques.

Other tests unique to the researched phenomena and time series were performed, including:

- **Seasonality:** Seasonality is the occurrence of systematic, predictable, and periodic patterns or fluctuations in a time series at regular intervals such as weekly, monthly, or yearly. Natural variables such as weather and climate can cause these patterns, as can man-made influences like holidays and business cycles. Seasonality substantially impacts time series forecasting, and failing to account for it can result in erroneous predictions and poor performance.
- **Stationarity:** Stationarity is a time series property in which statistical variables such as mean, variance, and autocorrelation remain constant throughout time (Box, 2013). Because it is less impacted by external causes and exhibits more predictable patterns, a stationary time series is easier to model and forecast than a non-stationary one. The Augmented Dickey–Fuller (ADF) test and the Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test are two methods for testing stationarity.

- **Trends:** A trend in a time series is a long-term pattern or direction that can be upward (positive trend) or downward (negative trend) (Box, 2013). Trends can be driven by several variables, including changes in population, technology, or economic situations, and they can considerably impact time series forecasting. Identifying and modeling trends is crucial in time series forecasting, as they can significantly impact prediction accuracy and contribute to bias and errors.
- **Individual Conditional Expectation (ICE):** ICE plots visualize how a model's predictions change when a specific feature varies while keeping all other features constant. Unlike Partial Dependence Plots (PDP), ICE plots show instance-level effects, revealing heterogeneity and potential nonlinear relationships. They help identify feature interactions and operational thresholds that impact predictions. ICE is crucial for interpreting complex models in domains like energy forecasting and fault detection.

2.3. Proposed framework

Wind power forecasting is a challenging endeavor due to the intermittent and unpredictable nature of the wind. Weather conditions, geography, topography, and wind farm specifications must all be taken into account for accurate forecasting. ML algorithms have shown potential for predicting wind power by studying patterns and relationships in historical data.

The suggested framework is a three-tiered one. First, the dataset is loaded and pre-processed with the necessary analysis and missing data filling. It is important to note that the indexation process is a date-time one. Therefore, dropping columns changes the timeline. Moreover, the analysis provided by the lagged periods showed the interconnections between each sample and at least three of its lagged versions. This was kept in mind during the filling of the values as well. The use of an ML KNN model for the filling of the missing data is intentional; in fact, this model ensures that several k neighbors are taken into account while the imputation process is taking place. The preprocessed data is then fed into a loop of adjustable regressors, which can take any number of desired regressors. The choice of using a loop is because energy forecasting papers do not compare one model directly with another. Some papers consider ML models to be more accurate, while others prefer DL models. Using a training loop helps us to see how different models perform and determine which one is likely the best from the start. A three-sample window and a step of one (step = 1) are evaluated; therefore, three samples are used as basic information, and the fourth is predicted each time. This choice was not made at random; the fact that three lags (lag=3) demonstrated a strong association with the target variable in the PACF (see Fig. 8) motivated this choice. The outputs of each regressor are then kept in a dataset for later tier use, along with the metrics results for comparison. The second tier involves extracting a seasonality trend from the target variable and the pre-processed dataset, which is significant given that the data analysis revealed a seasonal and trending behavior. The Prophet framework is utilized for this purpose, with a time limit of one day, to achieve a better correlation with the first layer, and the results are then saved. In the third tier, the first results from the regression models are combined with the second session and trend features from Prophet to form a final dataset, which is used to train a final estimator for predicting generated wind power. The metrics results are then calculated and compared with the results of the first-tier singular model. The proposed framework is depicted in Fig. 1.

2.3.1. Models and metrics

The trained models depicted in Table 3 were chosen based on efficacy in forecasting renewable energy quantities in various areas. The training process was conducted on an HP notebook computer with 8 GB of RAM using Google Colab, a Jupyter Notebook cloud platform that enables users to write, run, and share Python code from any

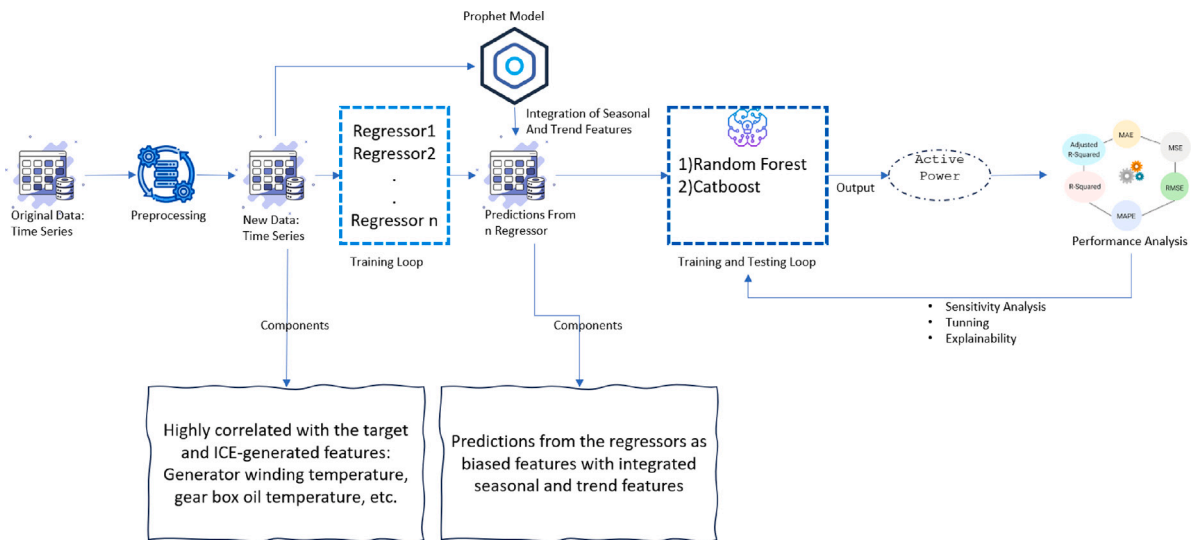


Fig. 1. The proposed framework.

location with an internet connection. It is a free Google service that gives users access to a GPU and TPU for the computation of large-scale machine-learning models.

The performance of the regressors was assessed using well-established regression metrics widely applied in this field, as evidenced by the literature. Specifically, Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the Coefficient of Determination (R^2) were utilized to evaluate the accuracy and reliability of the model predictions. These metrics provide a comprehensive understanding of how well the predicted values align with real-world data. Table 4 provides the used metrics, the mathematical expressions, and the limitations, where y_i and $\hat{y}_i$ represent the i th true and predicted values of the target variable, respectively, and n represents the number of samples to be predicted.

2.3.2. Final estimator choice

To determine the most suitable final model, an initial selection was made from the top-performing candidates, specifically Random Forest (RF) and CatBoost. Both models were first evaluated in a baseline configuration with randomly assigned parameters. Subsequently, a systematic hyperparameter optimization was conducted for each model to refine its performance. The selection process involved assessing the results from the optimized configurations and identifying the model that achieved the best performance. Notably, a comprehensive range of hyperparameters was explored to ensure thorough coverage of potential outcomes. Detailed information on the tuned parameters and their respective search ranges is provided in the appendix.

3. Results and discussion

The dataset was transferred to Google Colab in CSV format using the Python engine, facilitating streamlined analysis. Subsequently, a time indexation operation was applied to transform the dataset into a time series-indexed format, with data recorded at 15 min intervals. The objective of the analysis was to forecast wind active power for an hour ahead and assess the predictive performance for both a day and a week ahead, employing our most effective regressor. In this section, the obtained results are presented and discussed in the following order:

- Data Analysis Results.
- Models Performance Analysis.
- Sensitivity Analysis and Sliding Window Effect.
- Confidence Intervals and Explainable ML.
- Generalization Performance and Practical Considerations of the Framework

3.1. Data analysis results

The analysis of missing values revealed multiple series containing NaN values, as illustrated in Fig. 2. Two primary imputation methods were considered: interpolation and the iterative imputer-filling framework, which utilizes regressors to detect hidden patterns in the data and appropriately fill the missing values. Additionally, categorical data were numerically encoded; the only categorical series was the wind turbine generator number (WTG). When analyzing the dataset series, it became evident that the minimum value is typically 0. However, multiple maximum values were attained during the wind turbine work, necessitating a scaling operation to put all the series into the same range. This is critical since the implementation of regressors depends on the data input, and various ranges may exacerbate the interception of the target variable. Fig. 3 provides a statistical summary of the dataset components.

The correlation study used three different correlation methodologies (linear and nonlinear): Pearson, Spearman, and Kendall, to reveal both linear and nonlinear correlations between variables and goals. Three weakly correlated variables with the objective variable were identified by the three correlation approaches: wind direction, nacelle position, and hub temperature. Wind turbines are normally designed to capture wind power from different directions. Therefore, wind direction has no impact on the generated power, as shown in Fig. 4. Furthermore, wind turbines are typically placed based on knowledge of the area's climatic parameters, so wind direction is already considered. The nacelle position is another non-influencing factor on the generated wind power; this feature relates to the wind turbine section that houses the gearbox and generator. Finally, the hub temperature links the blades to the wind turbine. It is also worth noting that all three correlation approaches share the top 10 correlated features with the generated wind power, while the order of relevance varies. Fig. 4 depicts the correlation analysis.

Outlier analysis identified several anomalies in the turbine status and ambient temperature series, which were addressed using the z-score method. Understanding variations in the target variable over the years is crucial for assessing whether there is an upward trend in the turbines' power generation. From a predictive modeling and health management perspective, these variations can also indicate whether a turbine is operating optimally, experiencing faults, or requiring further diagnosis. Box plots for the three years of data (2018, 2019, and 2020) were generated (see Fig. 5), revealing a median shift toward higher power values. The distribution further indicates an increasing number of values exceeding the 300-kW threshold. Notably, while the

Table 3
ML model description.

Model	Description
Decision Tree	A decision tree starts at the root node, also known as the head node, and branches out to child nodes; the structure is essentially a tree that searches for solutions to issues based on constraints and solves problems quickly. Two of them are the Decision Tree Regressor and the Decision Tree Classifier.
Random Forest (RF)	The random forest is a non-parametric learning engine that does not need to be optimized or hybridized. It is an ensemble method that aggregates numerous weak predictors' predictions. However, it is rarely used for wind forecasting.
XGBoost	The XGBoost algorithm is an additive model composed of M decision trees that are used to predict target variables in supervised learning situations using training data. XGBoost's basic learner is a decision tree. Adding new base learners minimizes the error between the prediction values and the target. The final predictive values are the sum of all the base learners.
LightGBM	The Light Gradient Boosting Machine (LightGBM) is a new distributed gradient boosting architecture that uses a decision tree method. It has the benefits of fast computation speed and high robustness, and it can fully mine the hidden information of random sequences. For these reasons, LightGBM's utility in predicting power appears promising.
CatBoost	a recent gradient boosting algorithm developed by Yandex. It is designed to operate with categorical data and possesses several unique properties that make it particularly useful for machine-learning applications. It handles categorical features using a symmetric tree structure and incorporates various techniques to prevent overfitting, such as random permutations of training data and gradient-based feature selection. Overall, CatBoost has demonstrated cutting-edge performance on various ML tasks and is well-suited for wind power forecasting.
SVR	Using a kernel function, the support vector machine approach maps input space to a high-dimensional feature space. It determines the nonlinear relationship between input and output variables. The model's generalization ability can be increased by reducing the structure risk, resulting in a strong statistical law with fewer input samples. Support vector regression has the advantage of solving nonlinear problems and achieving the global best solution.
KNN	The KNN method discovers the K nearest examples in a training dataset and classifies these K instances as candidate classes. Then, using the similarity between them as a weight, substitute the predefined threshold value to determine the sample's classification.
MLP	is a type of multi-layered perceptron. ANN maps input data to output space using simple forward-back propagation and supervised learning. It has been employed in a variety of research projects involving hydrogen and bio-diesel creation, as well as wastewater treatment and biogas production
Stacking	This algorithm works as ensemble learning by combining numerous prediction models into one that functions on various levels; it introduces the concept of meta-learning. In addition to our models' numerous forecasts, we do a final investigation to determine where the models were correct and where they were incorrect. We propose reducing generalization errors by using this strategy to reduce generalization bias.
Voting	The Voting ensemble model combines the predictions of multiple individual models to make a final prediction. It aggregates the predictions from each model and selects the majority vote or averages the predicted probabilities to determine the final prediction.
Bagging	The ensemble model (Bootstrap Aggregating) involves training numerous distinct models in different subsets of the training data. The subsets are generated by randomly sampling the training data with replacement (bootstrap sampling). Each model is trained independently on its own subset, and their predictions are pooled by average or voting to yield the final forecast.

interquartile ranges and medians remain relatively consistent across the three years, some differences emerge: 2019 exhibits a slightly higher median and a broader distribution, whereas 2018 presents more outliers. Considering potential influencing factors, such as operational adjustments and external conditions, would enhance the interpretation of these findings.

Working with time series data presents challenges in comprehending various phenomena, such as seasonality and trends. Uncovering hidden patterns within the data, resulting from consensus on daily, weekly, and monthly changes, is crucial in constructing robust forecasting models. In this study, we analyzed by plotting and examining the relationship between each value and its one-sided delay for both the active power and wind speed series, as depicted in Fig. 6. The choice of these variables for analysis stems from the clear understanding that wind speed is the first external phenomenon that significantly influences the generated active power; this is further validated by correlation analysis as illustrated in Fig. 4. Hence, gaining insights into the behavior of these series is pivotal in comprehending the overall power generation dynamics. Although the interpretation of these lagged points is difficult, it is clear that clusters can be generated for the daily

lagged version, one of which is at the bottom left of the plot Fig. 6, corresponding to values between 0 and 750 kW, then values between 750 and 1250 are randomly distributed across the diagonal, along with another cluster for values above 1250. The same observation can be seen for wind speed values, where a big cluster also emerges in the bottom left of the plot Fig. 6-(b) for values between 0.1 and 0.3. When the weekly and monthly lagged numbers are visualized, the same observation is made, and a linear-like relationship emerges.

The findings of the lagged periods, coupled with the wind speed and generated wind power, indicate that possible seasonal and trend elements are associated with the prediction of the target. Data decomposition into trend, seasonal behavior, and residual analysis is given in Fig. 7, which reveals an increasing generation trend over the years up to 2019, suggesting that long-term modeling can provide meaningful insights into the evolving dynamics of wind resources. Regarding the seasonal component, the analysis reveals repetitive and predictable patterns, indicating that past wind behaviors can be leveraged to forecast future conditions. For example, the similarities observed between the periods 2018–01 to 2018–10 and 2019–04 to 2019–10 illustrate the

Table 4
Regression metrics and their limitations.

Metric	Formula	Limitations
MSE	$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$	Mean Squared Error (MSE): Highly sensitive to outliers, which can disproportionately affect the result. It may not provide an intuitive interpretation without further context or visualization.
RMSE	$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$	Root Mean Squared Error (RMSE): While it provides a clear indication of error magnitude, RMSE is also sensitive to outliers and may not provide a straightforward interpretation in certain contexts.
MAPE	$MAPE(y, \hat{y}) = \frac{1}{n} \sum_{i=1}^n \frac{ y_i - \hat{y}_i }{\max(\epsilon, y_i)}$	Mean Absolute Percentage Error (MAPE): Sensitive to values near zero, which can lead to significantly distorted percentage errors. This metric becomes unreliable when the actual values are close to zero.
R^2	$R^2(y, \hat{y}) = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$	Coefficient of Determination (R^2): While it provides a measure of goodness of fit, R^2 depends on the number of predictors and can fail to capture complex nonlinear relationships between the features and the target variable.
MAE	$MAE = \frac{1}{n} \sum_{i=1}^n y_i - \hat{y}_i $	Mean Absolute Error (MAE): Unlike other metrics, MAE treats all errors equally and does not differentiate between larger and smaller errors, potentially masking the impact of significant prediction errors.

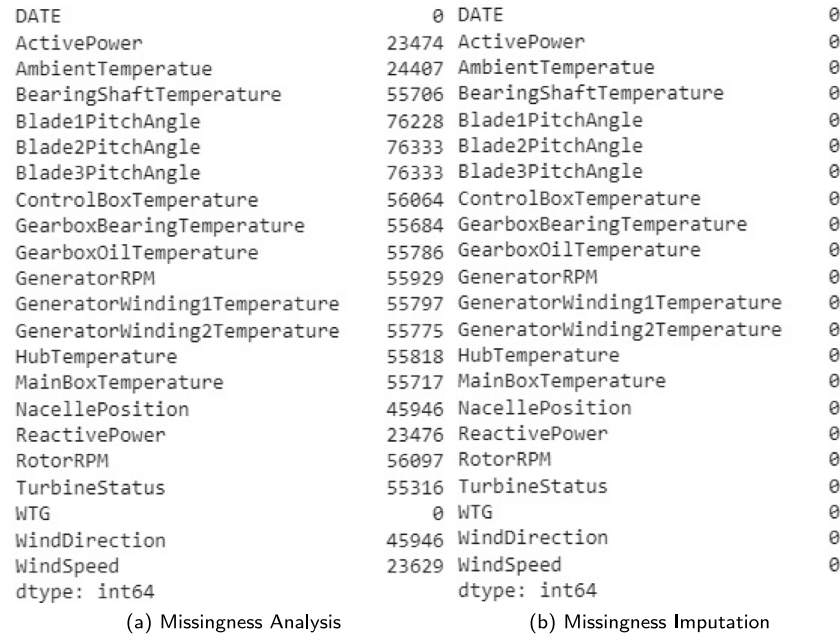


Fig. 2. Imputation process using iterative imputer and interpolation: (a) before imputation and (b) after imputation.

potential of seasonality-based forecasting. This characteristic is particularly valuable for machine learning models, as it enables the integration of historical seasonal structures to improve prediction accuracy.

The autocorrelation and partial autocorrelation plots in Fig. 8 show that the target wind power series is correlated with at least the first three lagged versions (lag=1, lag=2, lag=3), with a strong correlation for the first and a moderate correlation for the other two. Additionally, the lag=5 and lag=6 versions have a weak correlation with it. The autocorrelation function results indicate that models such as ARIMA and SARIMA may be beneficial in forecasting the target variable or, at the very least, eliminate the link between time series. Unfortunately, a more extensive discussion of monthly or weekly dependencies is limited. This constraint arises because each resampling step reduces the dataset size, thereby decreasing both the amount of training data available and the confidence in the forecasting results. Consequently, the intervals identified in the plots provide a measure of confidence in

the predicted outcomes, as these limited historically linked values help calibrate and guide the model's forecasts.

The ICE plot in Fig. 9 provides a detailed visualization of how ActivePower responds to variations in the four most correlated features: GeneratorWinding1Temperature, GeneratorWinding2Temperature, GeneratorRPM, and GearboxOilTemperature. These features were selected based on the correlation analysis, which identified them as having the strongest relationships with ActivePower. The ICE curves for Generator Winding Temperatures show a generally increasing trend, with noticeable plateaus and sharp upward jumps. This suggests that higher generator winding temperatures correspond to greater ActivePower output, likely due to the increased thermal load generated during power production. The presence of stepped transitions indicates operational thresholds where power remains relatively stable until a specific temperature range is reached, triggering a significant increase. The relationship between GeneratorRPM and ActivePower

	mean	std	min	25%	50%	75%	max
AmbientTemperature	28.789143	3.892203	0.000000	26.350811	28.844826	30.714347	4.240560e+01
BearingShaftTemperature	22.744205	21.844877	0.000000	0.000000	36.097828	43.291578	5.508865e+01
Blade1PitchAngle	60.667572	39.746197	-43.156734	5.487225	88.719589	88.719589	9.014361e+01
Blade2PitchAngle	60.728741	39.444092	-26.443415	5.989932	88.548264	88.548264	9.001783e+01
Blade3PitchAngle	60.728741	39.444092	-26.443415	5.989932	88.548264	88.548264	9.001783e+01
ControlBoxTemperature	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000e+00
GearboxBearingTemperature	33.979607	32.952625	0.000000	0.000000	49.179184	65.446392	8.223793e+01
GearboxOilTemperature	55.805031	4.957088	0.000000	53.839439	53.839439	57.500771	7.076458e+01
GeneratorRPM	580.683533	670.574402	0.000000	0.000000	30.771111	1155.511871	1.809942e+03
GeneratorWinding1Temperature	67.631851	17.217482	0.000000	62.229546	62.229546	67.010172	1.267730e+02
GeneratorWinding2Temperature	67.042603	17.222717	0.000000	61.686092	61.686092	66.257864	1.260430e+02
HubTemperature	38.360760	4.068040	0.000000	36.996185	39.996185	39.996185	4.799619e+01
MainBoxTemperature	20.909475	20.176540	0.000000	0.000000	31.711111	39.832761	5.425000e+01
NacellePosition	189.182190	69.612030	0.000000	172.000000	178.000000	195.500000	3.570000e+02
ReactivePower	70.633011	110.142548	-203.182587	-0.068410	0.660532	108.893642	4.037136e+02
RotorRPM	5.206415	6.014602	0.000000	0.000000	0.130289	10.362055	1.627349e+01
TurbineStatus	1214.371216	261587.500000	0.000000	2.000000	2.000000	2.000000	6.574653e+07
WTG	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000e+00
WindDirection	189.182190	69.612030	0.000000	172.000000	178.000000	195.500000	3.570000e+02
WindSpeed	6.705475	2.867648	0.000000	4.285497	6.422465	10.014299	2.297089e+01
ActivePower	496.182251	600.383423	-38.524658	0.000000	228.051064	838.515244	1.779032e+03

Fig. 3. Statistical insight on the dataset.

is also strongly positive, demonstrating a clear and predictable effect where an increase in RPM results in higher energy generation. The ICE curves for RPM show a smooth and continuous rise, particularly at higher values, indicating that ActivePower is highly dependent on rotational speed. The steeper slope at high RPM levels suggests that power efficiency gains become more pronounced beyond a certain speed threshold, reinforcing the importance of maintaining optimal rotational speeds for maximum power output. In contrast, the ICE plot for GearboxOilTemperature exhibits a less structured pattern. Many of the ICE curves remain flat, suggesting that ActivePower is not consistently affected by changes in oil temperature. However, at specific temperature points, sudden increases in ActivePower are evident, potentially indicating critical lubrication conditions where temperature variations influence turbine efficiency. The correlation analysis provided a statistical measure of association between these features and ActivePower, but ICE plots offer a deeper, instance-level interpretation of these relationships. While correlation captures global dependencies, ICE plots highlight local variations, revealing nonlinear behaviors and operational thresholds that might not be evident in correlation coefficients alone. This is particularly useful for understanding complex energy systems where multiple variables interact dynamically. The insights from the ICE analysis suggest that power generation models should account for nonlinear effects and potential interaction terms rather than assuming simple linear dependencies. This reinforces the importance of advanced interpretability techniques in energy system modeling, enabling more informed decision-making and optimization strategies for enhancing efficiency and reliability.

We conclude from the analysis that seasonal and trending factors influence this target variable and that its possible breakdown into these components could lead to a better understanding of its changes, thereby improving the regressors' prediction outcomes. This step was

performed using Prophet, a method for forecasting time series data based on an additive model that incorporates yearly, monthly, and daily seasonality, as well as holiday impacts. It works best with time series with substantial seasonal effects and data from multiple seasons. In addition, Prophet is resistant to missing data and trend alterations, which is helpful in this case, and it typically handles outliers well. Moreover, a strategy employing data windows was developed and implemented within a dedicated function to analyze the autocorrelation and correlation of lagged versions with the target variable. The primary objective of this function is to facilitate the adoption of a forecasting horizon based on relative choice, allowing for the testing of the solution across multiple future time intervals. In addition, the function illustrates the target variable's dependence on its lagged versions. Its parameters include the dataset, the time steps, and the window size, and it returns a dataset consisting of windows formed with a target variable based on time steps. This approach systematically examines the target variable's behavior over various temporal contexts, enhancing the forecasting process and providing valuable insights into its time-lagged relationships.

3.2. Model performance analysis

- **Preliminary Results:** The first tier of the training process consisted of a loop of 7 regressors: DT, RF, XGB, LGBM, MLP, stacking, and voting. Other models, such as KNN and GBR Ada tree, were excluded for poor results. The five regression metrics in Table 4 were considered. The resulting evaluation values are shown in Table 5.

The regression metrics values were broken down into a bar graph in Fig. 10 to visualize the best performing model. The RMSE is divided by 100 in the plot to allow us to see all the metrics

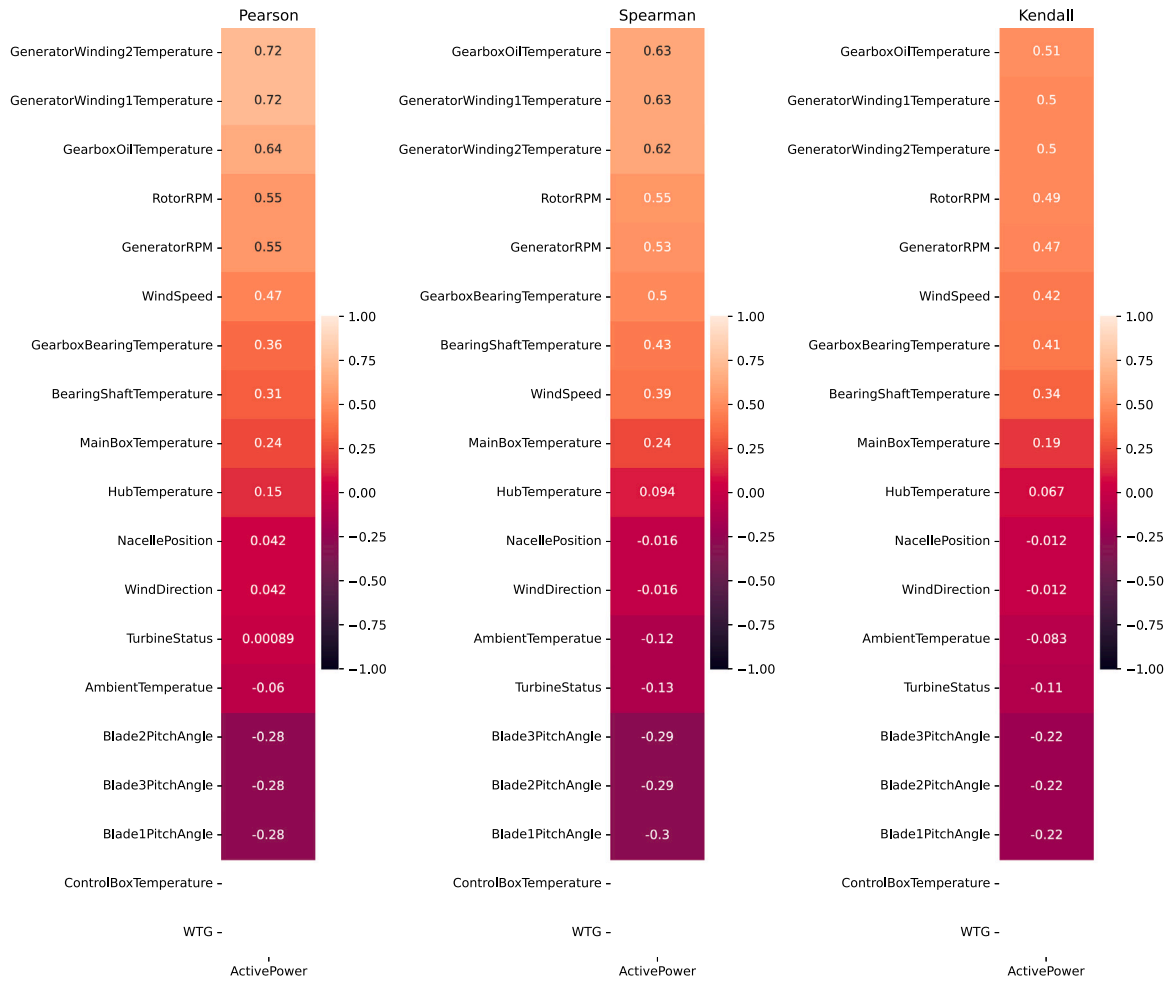


Fig. 4. Correlation analysis between generated power and collected features.

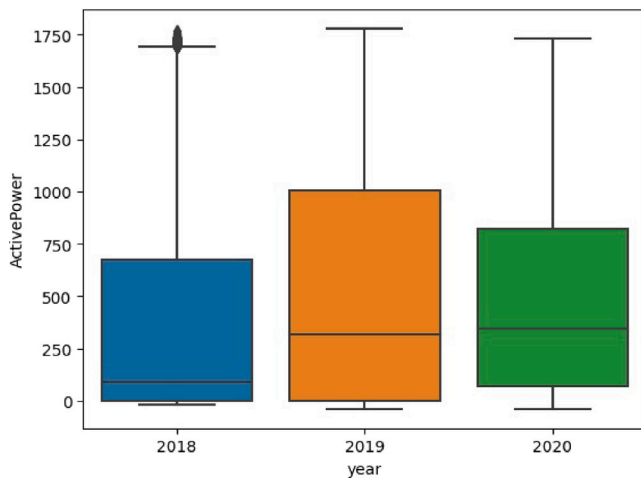


Fig. 5. Generated power distribution across three years.

simultaneously. The findings reveal that the random forest regressor is the highest-performing method, followed by the stacking ensemble model and the voting from the same category, with the best RMSE value of 75 and an R^2 result up to 0.99 rounded. The line plot of actual versus fitted values in Fig. 11 demonstrates that the regressor can capture temporal and target dependencies and that the predictions are well suited for values between 0 and

1000 kW but fail to follow the target for values above 1000 kW. This demonstrates that the regressor is poor at predicting large values; nonetheless, it effectively follows the target variable. The difficulties in predicting values beyond 1000 kW can be attributed to outliers, seasonality, and trend handling. In reality, the latter could impact forecasting accuracy if the model cannot catch these patterns, resulting in regular cycles not being caught. The trend can have a comparable effect on predicting because it is linked to long-term upward or downward trends in data beyond short-term fluctuations. Therefore, the model may not capture the general direction and amplitude of the trend, resulting in inaccurate long-term forecasting.

• **Full Tiers Results:** The results of the first layer were saved into a new dataset, which was then sent through the Prophet model to extract trend and seasonality aspects from its target variable. These features were then combined with the results of the first-tier process. Indeed, these results do not all have the same accuracy, but this diversity allows our final estimator to learn from the faults of the previous ones; additionally, selecting only the best regressors may result in overfitting. Because each dataset sample includes the prediction of several regressors across a specific window, the link with the first layer allows us to treat the target variable appropriately. The CatBoost framework was used as the final estimator, and the prediction results were compared to the best random forest results in Fig. 12. It is important to note that the comparison presented here focuses exclusively on three key metrics: R^2 , RMSE, and MAE. These metrics are widely recognized and commonly used in the literature for evaluating

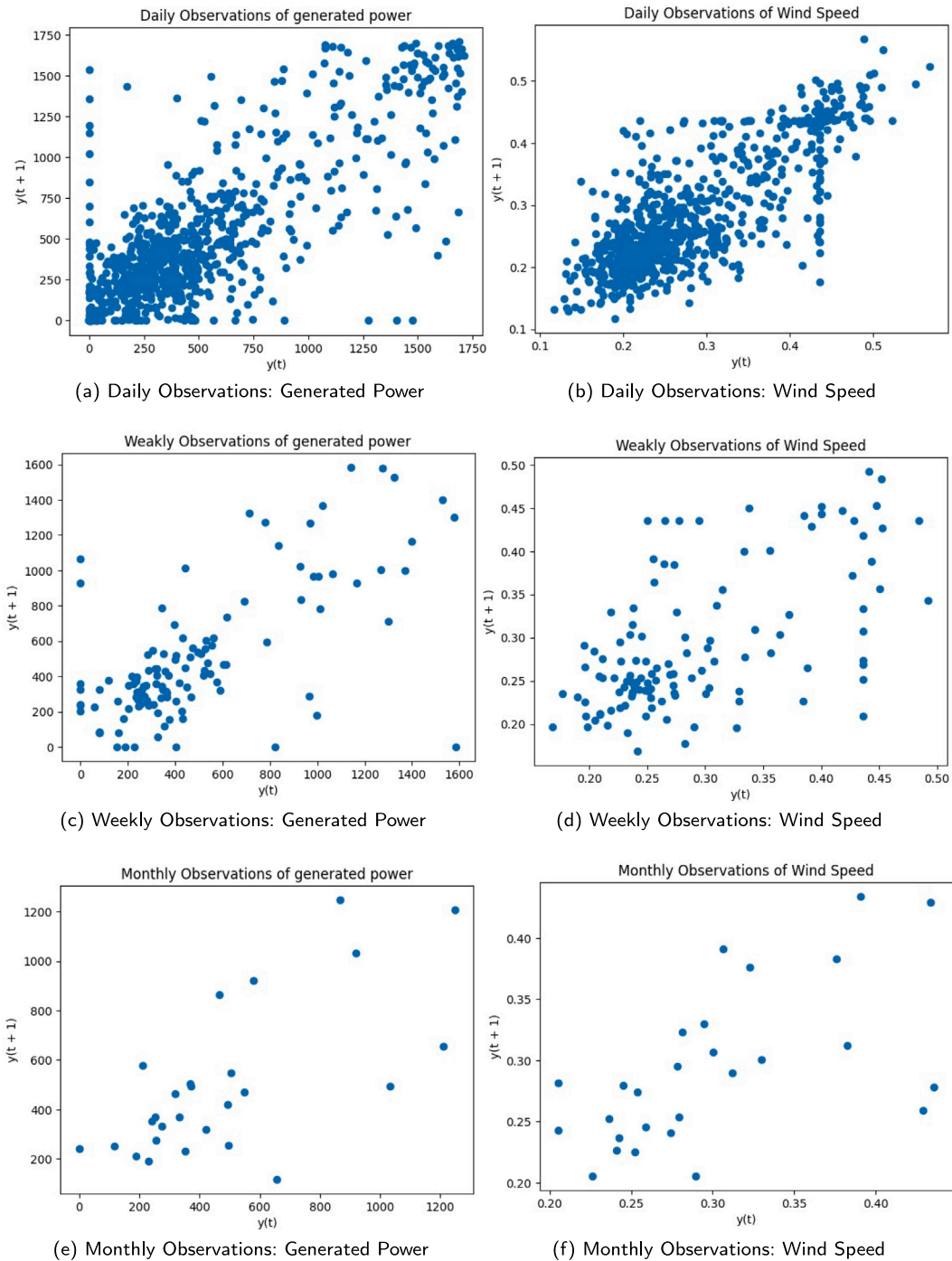


Fig. 6. Daily, weekly, and monthly lagged relationships.

model performance. The findings show that the suggested hybrid model has an RMSE of 30.60 for short-term (15 min) forecasting and an RMSE of 50 for forecasting one day ahead. At the same time, it drops to 41 for 1 week ahead of forecasting, which is at least half the RMSE of the best regressor in the individual prediction procedure. In addition, the hybrid model outperforms the solo learners in all regression metrics, confirming its superiority. To see the difference in prediction and better grasp the hybrid model's superiority over the others, the plots of actual vs. fitted values and the residual plots of both RF and hybrid (Prophet + CatBoost) are shown in Figs. 13 and 14. Looking at the results, the predictions appear to improve early on; nevertheless, the most substantial optimization happens in the late phases when

viewing the predictions above the 1000 kW barrier. This insight is also evident in the residual plots, illustrating how the difference between the real and fitted values is near zero. It can be seen that more data for the hybrid model is centered around zero and that the difference in residual achieves a value between -200 and 200 , but it increases to the 1000's for the RF model. Indeed, trend and seasonality analysis improved prediction outcomes; the suggested model considers diurnal patterns, presented in Elliott et al. (2004) as wind changes that might occur throughout the year and depend on seasonal behavior. Moreover, combining numerous regressor predictions improved the final estimator's predictive ability.

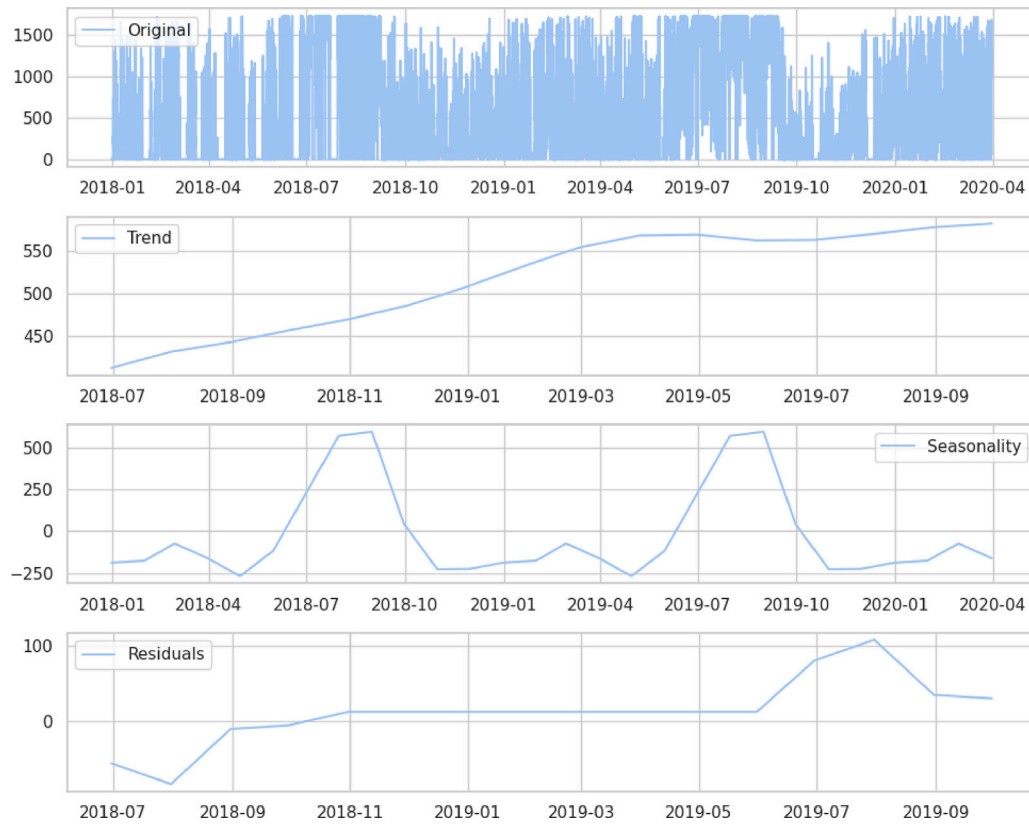


Fig. 7. Decomposition plots for the predicted variable.

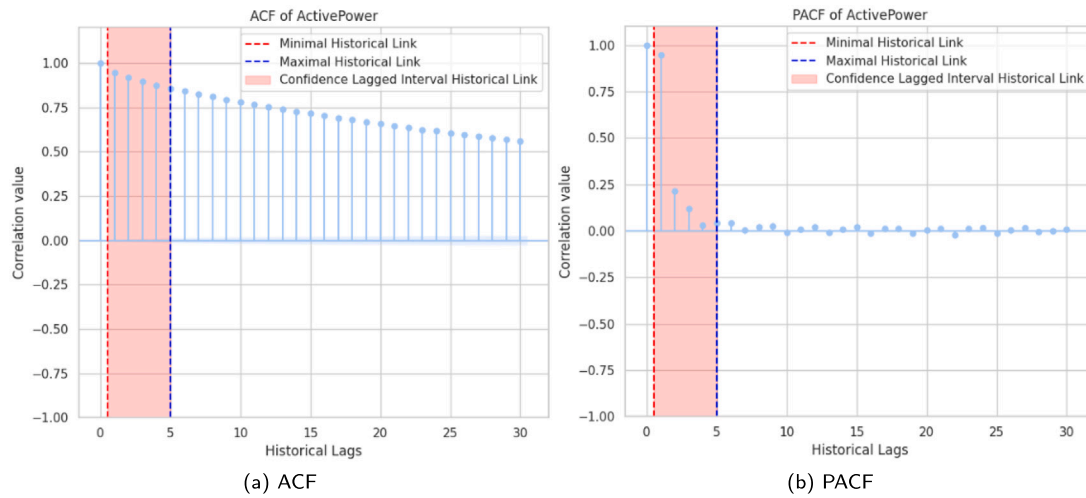


Fig. 8. Autocorrelation and partial autocorrelation plots: (a) ACF and (b) PACF.

Table 5
Regression metrics results.

Algorithm	MAE	MSE	RMSE	RSE	RAE	Median Absolute Error	R^2
Random Forest	17.116	5703.221	75.520	0.022	0.041	3.423	0.989
Decision Tree	18.829	9014.247	94.943	0.035	0.045	4.813	0.975
XGB	25.347	10590.660	102.911	0.041	0.060	6.388	0.970
Light GBM	30.137	12990.897	113.978	0.051	0.072	7.389	0.951
MLPerceptron	102.821	30825.864	175.573	0.120	0.244	64.886	0.991
Stacking	20.251	7278.411	85.314	0.028	0.048	5.587	0.983
Voting	23.103	8658.096	93.049	0.034	0.055	5.009	0.977

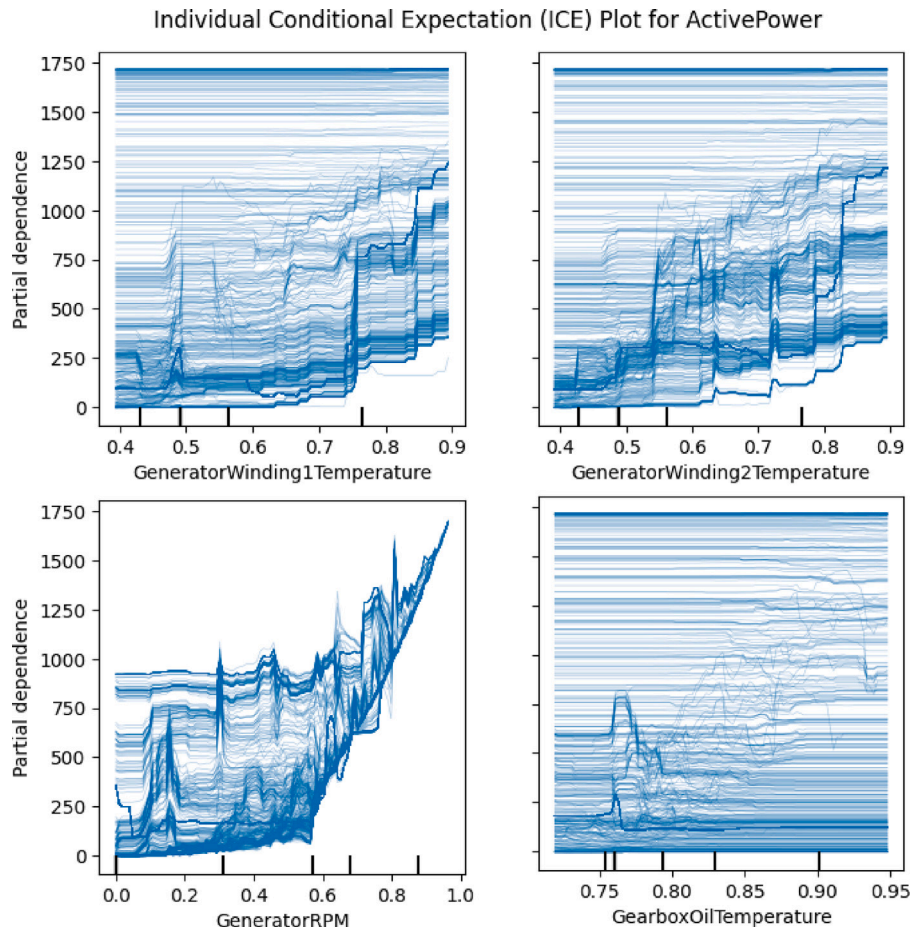


Fig. 9. Individual conditional expectation between the target and most correlated features.

3.3. Sensitivity analysis and sliding window effect

It is crucial to understand the interplay of various parameters in predicting the target. Parameters such as the sliding window and the resampling rate can significantly influence the model's decisions, as they affect the data and the batches fed to the regressor. A thorough analysis of both parameters is provided below.

- **Sliding Window Effect:** Based on the time-lag autocorrelation study, the three-step window suggests that the three prior lags of the predicted variable are associated to some extent with the current one. As the window increases, the observed findings reveal that the regressors perform poorly, as seen in Fig. 15. However, it is clear that at a window of 5, the whole system outperforms the first window choice of 3, with an RMSE of 27.92. This is because in the PACF plot Fig. 8, we discovered that each sample is at least correlated with its five-lag versions. Still, no discernible association is observed after that. The selection of a 10- or 15-sliding window validates this assumption since the system's overall behavior continues to worsen, and the RMSE observed for the last window examined is at least greater than the best one by 10.
- **Resampling Rate Effect and Forecasting Horizon:** Hybrid models outperformed all others in all prediction periods ranging from 15 min to one day and eventually one week. However, this does not change the fact that the value of the RMSE metric and all others grew as the forecast period lengthened, as shown in Fig. 15. In reality, this is explained by the fact that the data set in

question is a 15 min frequency dataset, and resampling from minutes to days reduces 113700 samples to just 730 data sets, and resampling to weeks reduces 113700 samples to only 109 data sets. Considering everything, including the fact that the more data there is, the better the accuracy, these results are not surprising. The best-performing models for the different forecasting periods are provided in Fig. 15.

- **Hyperparameter Tuning:** Hyperparameter tuning was performed within the ranges specified in the appendix, focusing only on the final estimators to determine the best configurations. The results are presented in Fig. 16, where the performance of different models was evaluated using key regression metrics: RMSE, MAE, and R^2 (%). RF model, depicted in Fig. 16(a), demonstrates a stable and well-structured optimization process. The RMSE values range between 80 and 100, showing minimal fluctuations across optimization rounds. Similarly, the R^2 score starts at approximately 78% and steadily improves to around 98%, indicating a consistent increase in the model's ability to explain variance in the target variable. The MAE also follows a stable pattern, remaining between 20 and 30 with minor deviations. These trends suggest that the tuning process for the RF model was effective in identifying hyperparameters that gradually enhanced performance without introducing erratic behavior. The absence of extreme peaks or sudden drops in the metrics confirms that the optimization converged toward a reliable configuration. In contrast, Fig. 16(b), which represents the performance of the second model, shows significant instability. The RMSE varies dramatically between 90 and 220, with frequent spikes reaching the upper limit. The MAE exhibits similar erratic behavior, fluctuating between 20 and 180, indicating that certain hyperparameter combinations

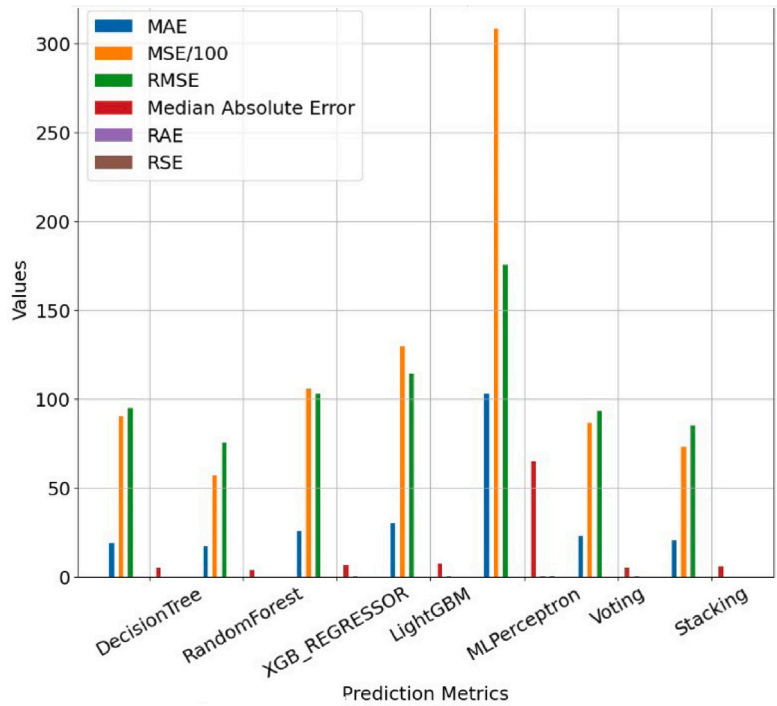


Fig. 10. Regression error metrics versus listed ML models.

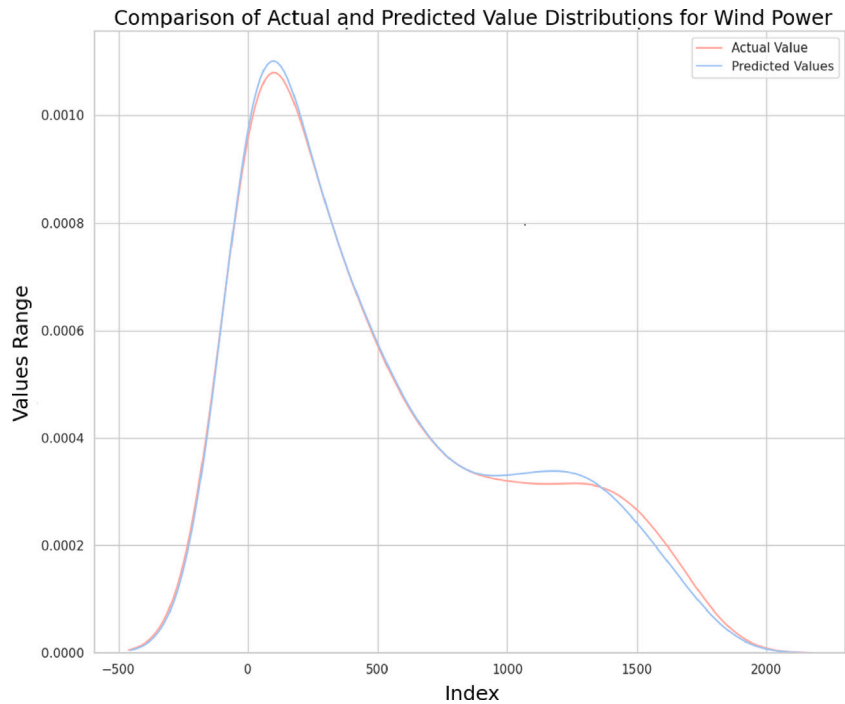


Fig. 11. Actual vs. predicted values.

resulted in considerable errors. The R^2 score, which ideally should be maximized, lacks a clear pattern, oscillating between 10% and 99%. Such instability suggests that the model's performance was highly sensitive to hyperparameter changes, leading to inconsistent learning behavior. This variation implies that either the search space was too broad or that some hyperparameters had a disproportionate impact on the model's predictions, causing drastic performance changes in certain rounds.

Comparing both tuning processes, it is evident that the RF model benefits from a more controlled and predictable optimization trajectory. The steady improvement in its RMSE and R^2 values suggests that the tuning process was effective in refining the hyperparameters to enhance performance. On the other hand, the second model exhibits chaotic behavior, making it less suitable for deployment without further refinement. To mitigate such instability, a more structured search approach can be implemented to

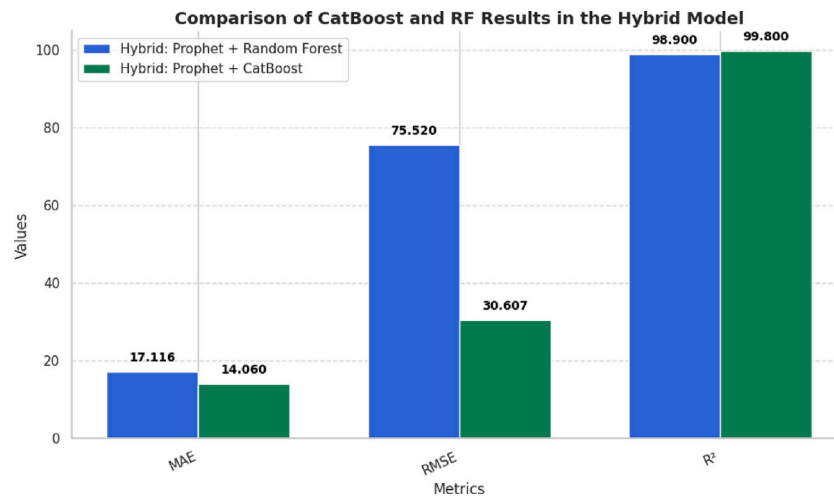


Fig. 12. Comparison of CatBoost and RF results as final estimators for the hybrid model.

navigate the hyperparameter space more efficiently and prevent extreme fluctuations.

3.4. Confidence intervals and explainable ML

Regression model prediction intervals serve to estimate a range of values within which future observations are expected to fall. These intervals play a pivotal role in decision-making by quantifying the uncertainty surrounding individual predictions rather than population parameters. In this study, the calculation and visualization of prediction intervals were performed for the optimal regression model. A novel 3-dimensional representation incorporated upper and lower bounds, as well as true and predicted values. By assessing the width of these intervals, we gauged the model's ability to capture data dependencies while accounting for inherent uncertainty. The residual standard error (RSE) metric was used to calculate these intervals, with the formula: $PI = (\hat{Y}) \pm t_{\alpha/2, n-2} * RSE$. It can be seen in Fig. 17 that the lower and upper bounds of the prediction intervals intercept all the true values; in addition, the predicted target is close to the area of the interval. However, the model is still struggling to find the target. An RMSE of 27 for the optimal window value is still significant and needs to be enhanced. The current dataset does not contain any information about the type of wind turbine, radius, or height, and can be viewed as a general case. These pieces of information are crucial in building modeling equations for the prediction system, thus enhancing the results. A scatter plot is generated for the best-performing model in Fig. 17(b). The results show that the predicted values closely align with the actual observations, with a near-linear relationship becoming especially evident after the first 500 samples. The time-series plots in Fig. 17(c) further emphasize the strong predictive capability of the model, although slight distortions are observed at specific intervals, notably between samples 370 and 380. These results demonstrate the model's robustness and its strong ability to capture and track the variations inherent in wind power generation data.

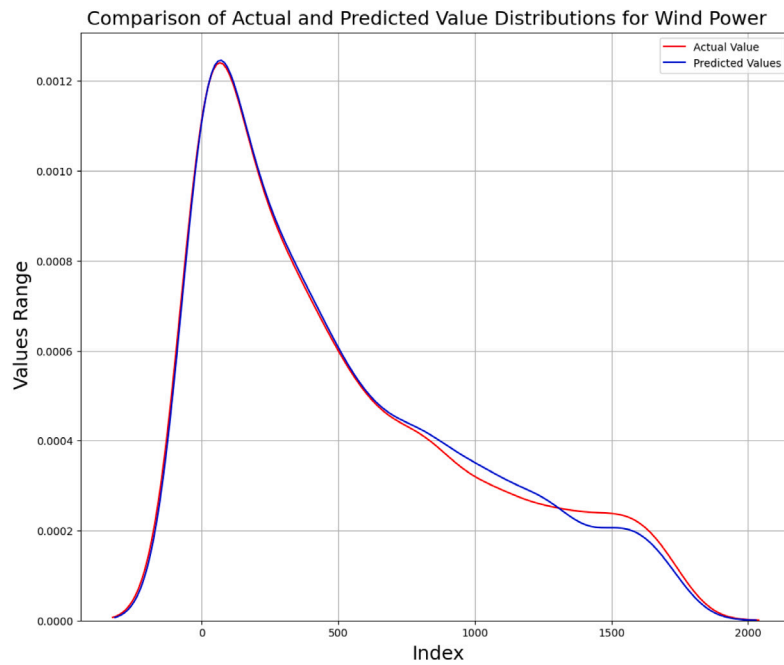
The ML model's explainability is crucial for unraveling the decision-making processes hidden inside the prediction algorithm. Various tools are available to achieve this goal, offering valuable insights into the factors influencing the model's decisions. SHAP (SHapley Additive exPlanations) is one such method that explains ML predictions by quantifying each feature's contribution to the model's output, facilitating a deeper understanding of the model's behavior. SHAP is of utmost importance in energy domains as it enhances model interpretability, aids in anomaly detection, validates model performance, provides policy insights, supports feature engineering, and fosters trust and ethical usage of AI in energy applications. In this case, the SHAP

explanation was considered for the final estimator for the hourly and daily predictions in Fig. 18. In hourly predictions, the influence of trend and seasonality features on the model's decisions appears to be limited. The interception of seasonality and trend from hourly data can be challenging due to noise and short-term fluctuations in the data. The granularity of hourly data contributes to more random variations, making it difficult to discern the underlying seasonal patterns.

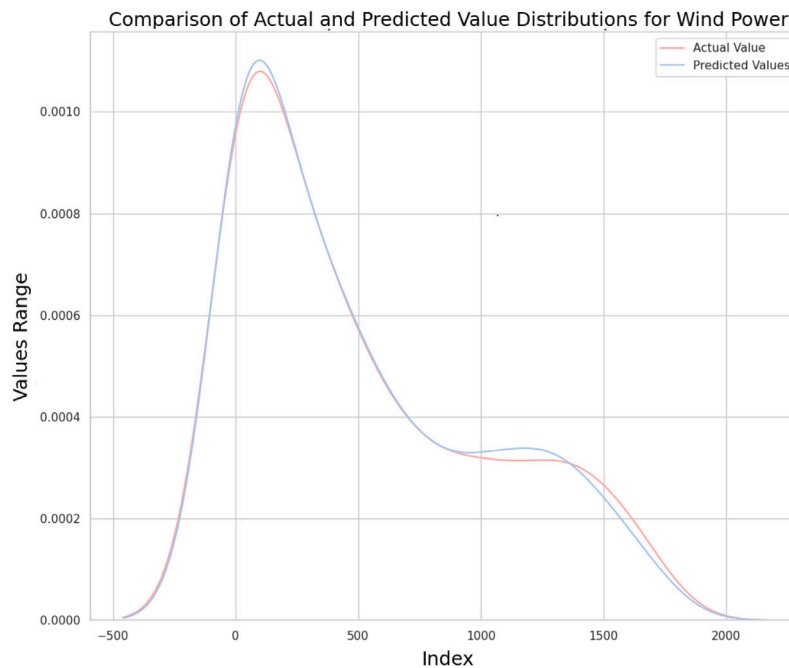
In contrast, a clearer picture emerges for time series with daily predictions. Daily interpretations of SHAP show that both the trend and 'y_hat_upper' feature gain more influence on the model's decisions. Daily data tends to smooth out short-term fluctuations and noise, revealing more prominent and stable patterns. This characteristic makes identifying patterns and interpreting seasonality easier, leading to a better understanding of the underlying factors driving the model's predictions. Furthermore, the final estimator leverages the strength of robust regressors to make decisive predictions. Notably, tree-based models and ensemble learners dominate in shaping the outcomes within the CatBoost framework. This hybrid system's primary objective was to harness the collective power of multiple regressors in a unified approach, thereby enhancing the overall predictive performance. By combining these diverse techniques, we achieve a more effective model. Another complementary methodology for assessing feature importance was employed using CatBoost's embedded permutation-based feature importance. While this approach is inherently tied to the model's own prediction mechanism, it offers valuable insight into how CatBoost achieves its superior performance. The results in Fig. 19 indicate that the trend feature consistently ranks among the top seven contributors to the model's predictions for both the one-hour and one-day horizons, albeit with a varying order of importance. Moreover, the analysis highlights that CatBoost, like other tree-based learners, strongly relies on recursive partitioning structures to calibrate predictions. Nonetheless, this does not preclude the potential benefits of hybridizing Prophet with more advanced neural architectures, which could further enrich predictive capacity, albeit at the cost of increased complexity and reduced interpretability.

3.5. Generalization performance and practical considerations

It is important to emphasize that, given the design of the proposed framework, its generalizability is straightforward. The acquired data undergoes preprocessing and is then passed through two stages: first, the extraction of trend and seasonality components, and second, the approximation of regressor-based representations of turbine readings. These outputs are then fed into the final estimator as generalized features. This methodology can be seamlessly adapted to other datasets,



(a) Random Forest line plot: true vs. predicted



(b) Hybrid model line plot: true vs. predicted

Fig. 13. Comparison of actual vs. fitted plots: (a) RF and (b) hybrid.

provided that sensor readings are available. Moreover, the framework's flexibility at the regression stage allows for the integration of alternative high-performing models, such as neural network architectures (e.g., LSTM), which are widely recognized for their effectiveness in time-series forecasting. The regressor selection process can be optimized either through systematic identification of the best-performing model or based on user-defined preferences. At the same time, Prophet demonstrates strong capabilities in capturing and modeling trend and seasonal components within the data. Following extensive experimentation and comparative analyses with both standalone learners and

hybrid configurations, the CatBoost–Prophet hybrid model was identified as the most effective, as detailed in Fig. 12. In contrast to many prior studies, which often partially report performance metrics, this work establishes a comprehensive benchmark baseline by systematically evaluating all major regression indicators, including RMSE, MAPE, MSE, MAE, R^2 , RAE, and RSE. Reporting a single metric in regression tasks can lead to incomplete or even misleading conclusions; therefore, the broad set of metrics presented here ensures a rigorous and transparent evaluation. Additionally, forecasting horizons are not always consistent across studies. For example, in Lin et al. (2021), long-term forecasts of 24–72 h ahead were conducted using

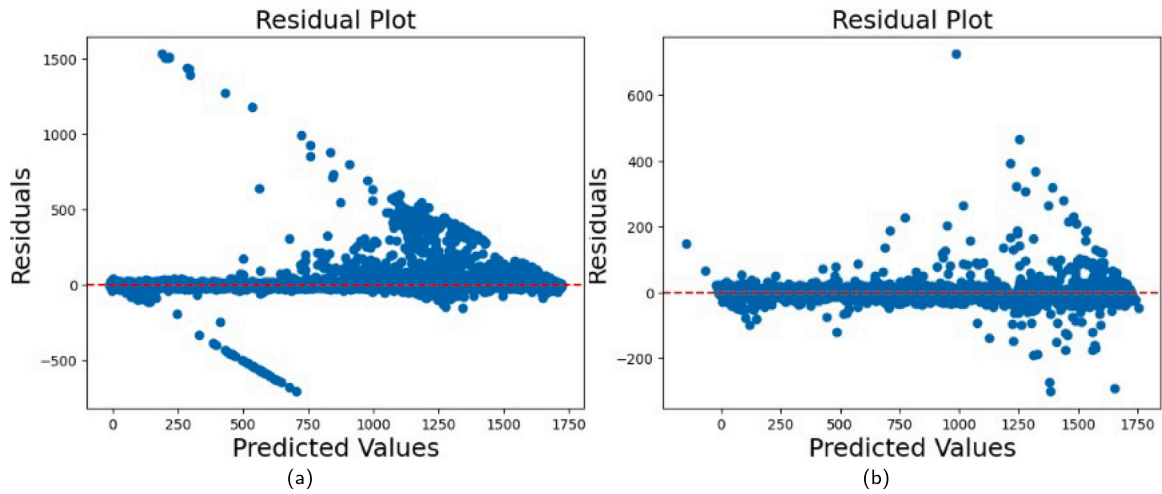
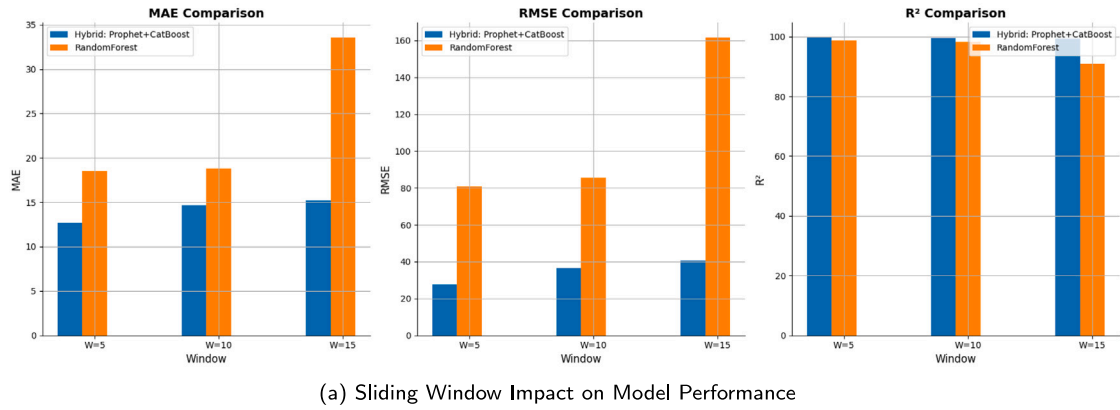
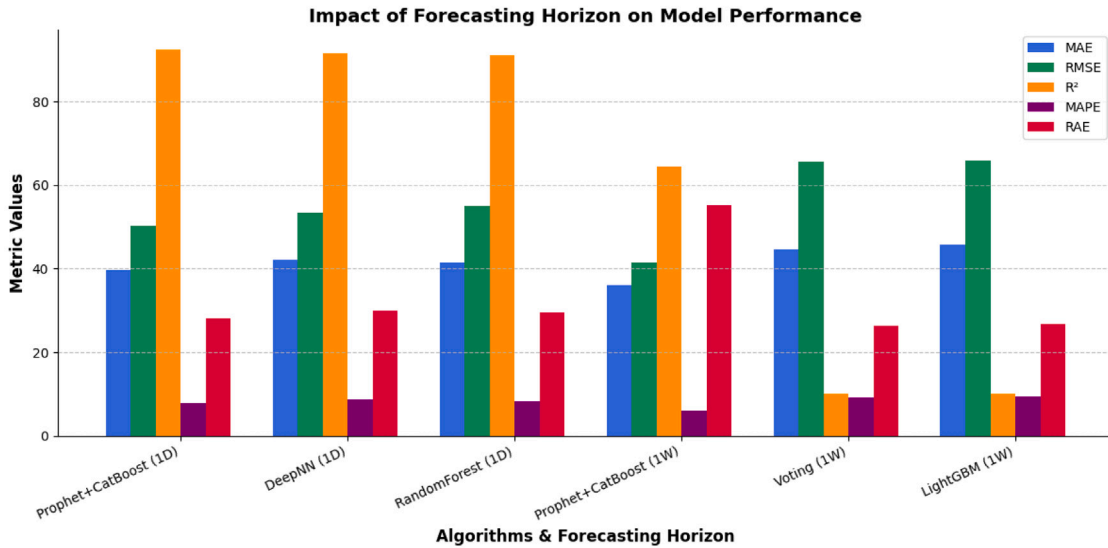


Fig. 14. Comparison of residuals plots: (a) RF and (b) hybrid.



(a) Sliding Window Impact on Model Performance

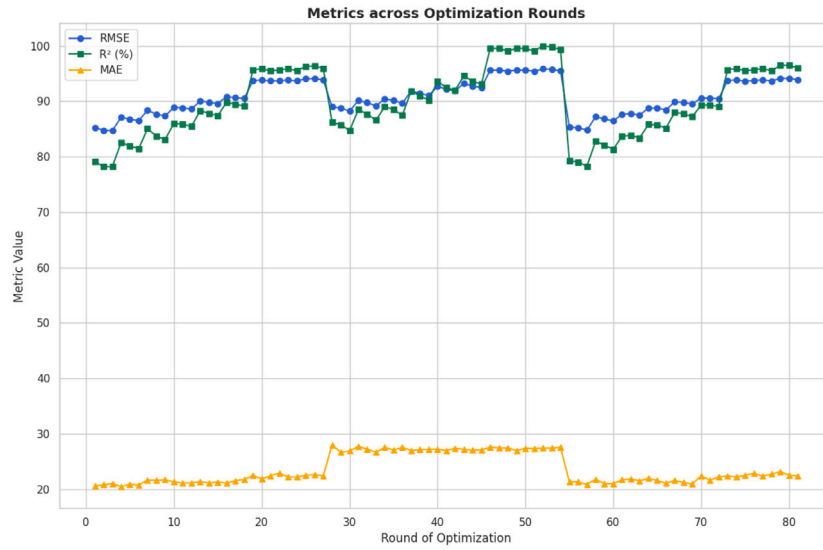


(b) Forecasting Horizon Impact on Model Performance

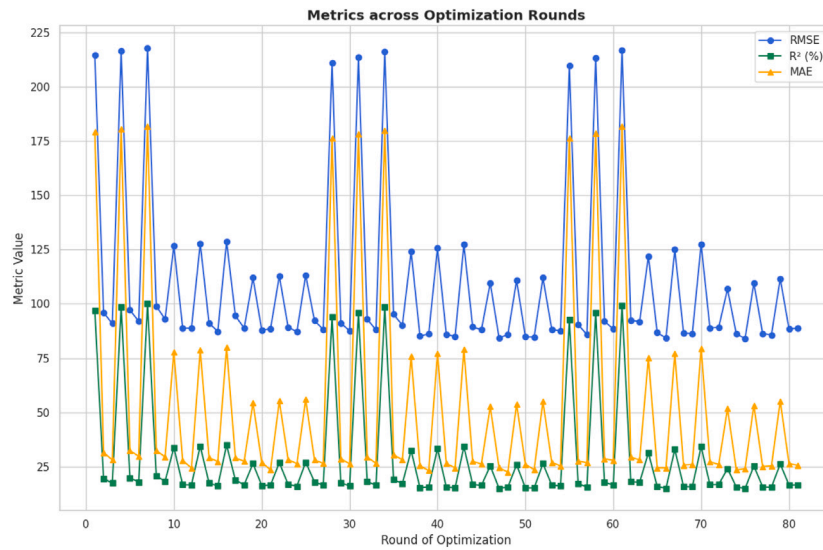
Fig. 15. Sensitivity analysis: (a) sliding window impact and (b) forecasting horizon impact.

Turkish SCADA wind power data, achieving a MAPE of 5.13% for the target horizon. In contrast, this work focused on short- to mid-term horizons, specifically 1-day ahead, 1-week ahead, and the original 15 min resolution, achieving MAPEs of 7.5% and 6%, respectively. It is important to note, however, that given the inherent variability of

wind patterns, longer forecasting horizons do not necessarily guarantee better or worse performance. The conducted sensitivity analysis further confirms that forecast accuracy is strongly influenced by multiple parameters, underscoring the complexity and volatility of wind power prediction. It is important to note that the observed increase



(a) Random Forest Tuning



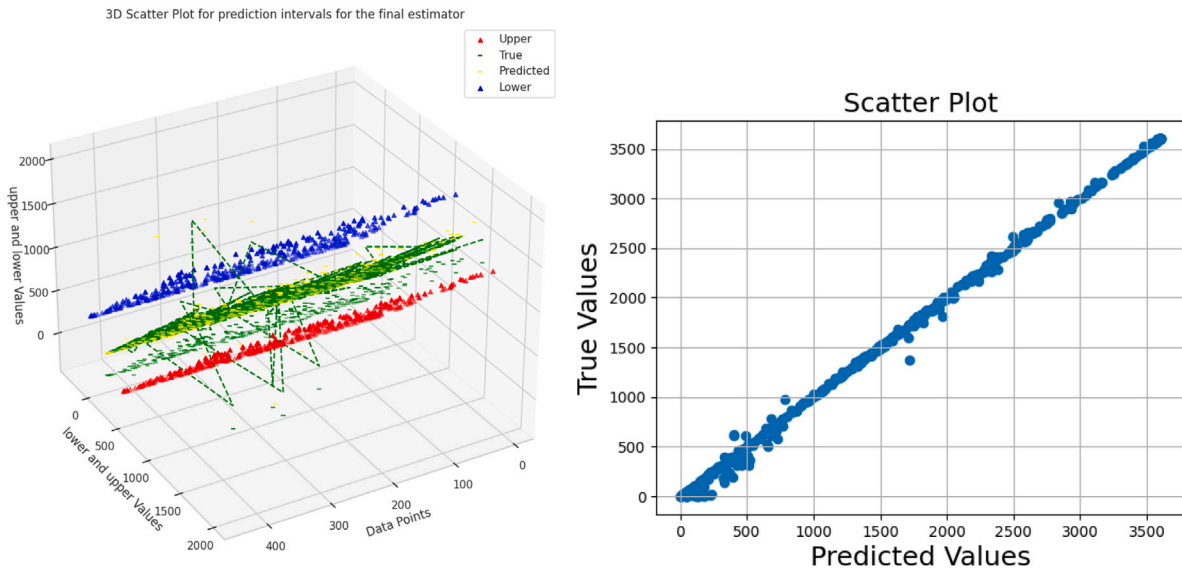
(b) CatBoost Tuning

Fig. 16. Final estimator hyperparameter tuning: (a) RF and (b) CatBoost.

in RMSE with longer forecasting horizons, as shown in Fig. 15b, is expected. This behavior arises from the inherent challenges associated with extending the forecast horizon and the corresponding growth in resampling intervals. The modeling framework is designed to capture the influence of critical lagged values of wind power on the current prediction, up to four data points as indicated by the PACF analysis. From a practical perspective, two key implications can be highlighted: first, increasing the availability of large data samples can mitigate the effects of shrinkage during resampling and improve the model's adaptability across different horizons. Second, while the model demonstrates promising performance for hourly forecasting, its practical utility can be further enhanced by integrating a validation mechanism that compares each prediction at $T + 1$ with the true observed value. This would enable online calibration, leading to more accurate future predictions. Although longer horizons could be tackled with more complex models such as LSTM, doing so would increase computational and memory requirements, which may limit deployment on small IoT devices. The current framework, by contrast, is optimized for efficiency and suitability for resource-constrained environments while maintaining robust forecasting performance. This comparison highlights the robustness

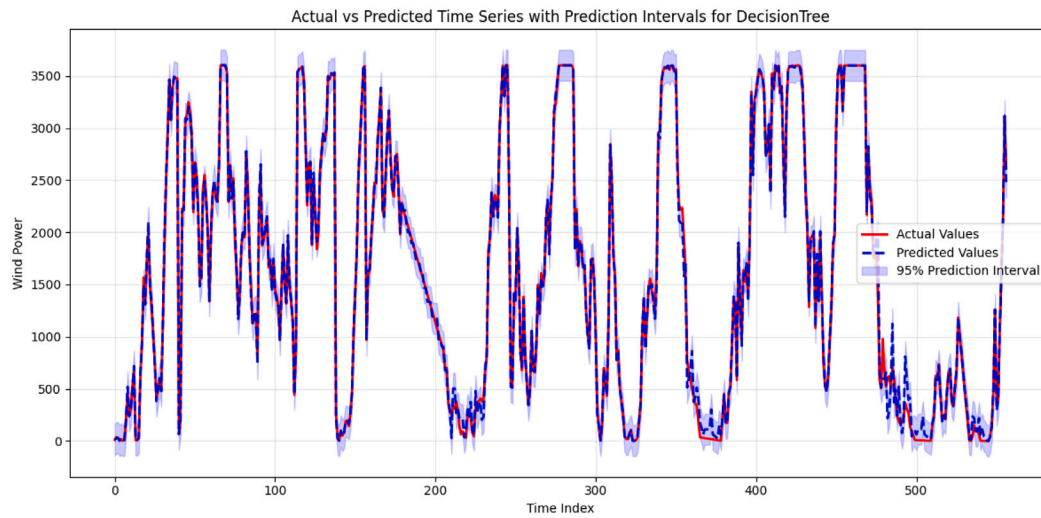
and adaptability of the proposed hybrid framework, which demonstrates competitive performance across multiple temporal scales while maintaining methodological transparency and generalizability.

Another critical consideration concerns the wind turbine parameters, such as hub height, rotor diameter, and manufacturer-provided power curves. In the proposed framework, the initial prediction stage explicitly integrates these parameters (as shown in Table 2) to estimate the corresponding wind power output. This ensures that the forecasts remain firmly grounded in the physical characteristics of the specific turbine under study. The subsequent stage, however, builds upon these predictions by combining regressor-based learning with trend and seasonal components extracted from Prophet. At this stage, the final outcomes depend primarily on the predictive capacity of the regressors rather than directly on turbine-specific input parameters. As a result, the framework exhibits strong theoretical generalization capability, allowing it to adapt its forecasts across turbines with varying configurations. With respect to site-related factors, none of the current input features explicitly capture location-specific conditions. Consequently, the model does not directly account for site variability. Nonetheless, when considering the wake effect, a phenomenon that significantly



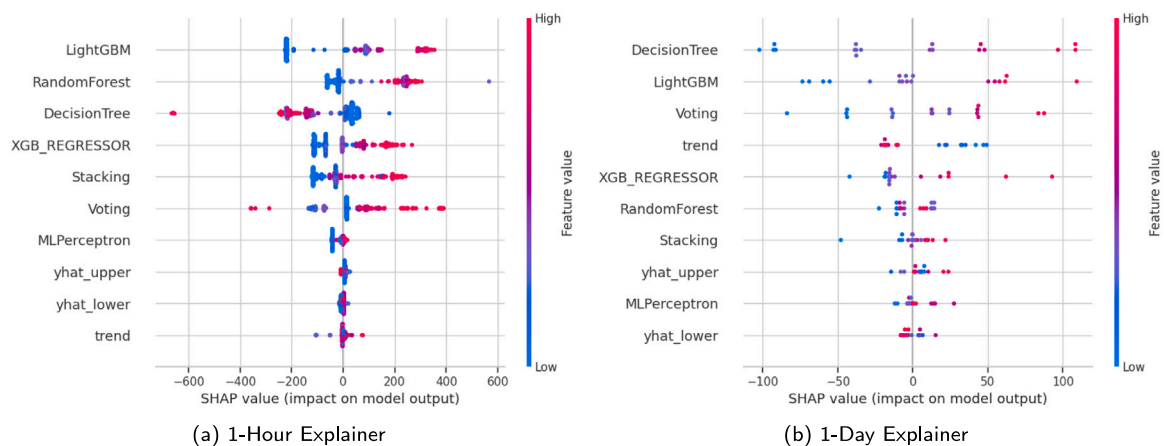
(a) 3D scatter for prediction intervals for the final estimator

(b) Scatter plot for the best model



(c) Time series plots for the final estimator

Fig. 17. Prediction intervals and model validation: (a) 3D confidence interval visualization, (b) actual vs. predicted scatter plot, (c) time-series overlay of actual and predicted values.



(a) 1-Hour Explainer

(b) 1-Day Explainer

Fig. 18. SHAP explanation results: (a) hourly predictions (1H) and (b) daily predictions (1D).

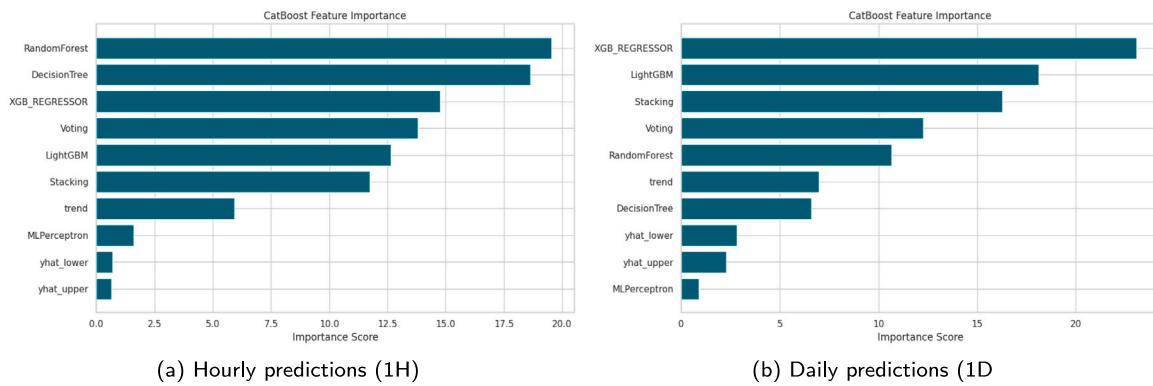


Fig. 19. Features importance of CatBoost: (a) hourly predictions (1H) and (b) daily predictions (1D).

influences downstream turbine performance, there is clear potential to extend the framework. By coupling the current predictive setup with established wake modeling approaches, such as the Jensen or Larsen models, or through computational fluid dynamics, it becomes possible to embed physical system constraints within the forecasting process. Such integration would enhance realism while preserving the framework's flexibility. Collectively, these aspects highlight the generalizable and extensible nature of the proposed hybrid solution.

4. Conclusion

In this study, data analysis, time series decomposition, and ML techniques were combined to develop a hybrid model for wind power forecasting. A dataset from a wind turbine in Turkey was acquired, analyzed, preprocessed, and utilized within a two-tier prediction framework. The first tier extracted trend and seasonality patterns from the data and integrated them into an initial dataset, which included random predictions from six regression models trained to estimate the default values of the target variable. This enriched dataset was then passed to the second tier, where predictions were refined using two calibration regressors: CatBoost and Random Forest. The results demonstrated that both final estimators significantly enhanced forecasting accuracy, achieving an R^2 value of 99.8% for 15 min interval predictions. However, the RMSE at the final stage remained at 30.607. Sensitivity analysis revealed that model performance declines as the forecasting horizon increases, which is expected due to the challenge of capturing long-term dependencies. Additionally, the originally used sliding window for data splitting was found to influence performance. While prediction intervals can be effective for short-term forecasting, the stochastic nature of wind makes it difficult to capture long-term variations accurately. The explainability analysis indicated that features introduced via Prophet improve forecasting, particularly over extended periods. To gain a clearer understanding of the framework's robustness, it is recommended to test it on larger datasets. Furthermore, replacing the final estimators with advanced models such as LSTMs or Transformers can potentially enhance prediction accuracy and improve long-term dependency modeling.

CRedit authorship contribution statement

Zaid Allal: Writing – original draft, Software, Methodology, Formal analysis, Data curation. **Hassan N. Noura:** Writing – original draft, Software, Methodology, Formal analysis, Data curation. **Ola Salman:** Writing – original draft, Software, Methodology, Formal analysis, Data curation. **Khaled Chahine:** Writing – review & editing, Validation, Supervision, Formal analysis, Conceptualization.

Table 6

Hyperparameter search space for CatBoost and Random Forest.

CatBoost	
Parameter	Search Range
Iterations	100 to 2000 (step=100)
Learning Rate	0.01 to 0.3 (log scale)
Depth	4 to 10
L2 Leaf Regularization	10^{-3} to 10 (log scale)
Bagging Temperature	0.0 to 1.0
Random Strength	0.0 to 1.0
Border Count	32 to 255
Loss Function	RMSE (fixed)
Random Forest	
Parameter	Search Range
Number of Estimators	100, 200, 300
Maximum Depth	None, 5, 10
Minimum Samples Split	2, 5, 10
Minimum Samples Leaf	1, 2, 4

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix. Hyperparameter tuning for the final estimators

This appendix presents a detailed overview of the hyperparameter tuning process for CatBoost and RF. For CatBoost, grid search cross-validation was employed to explore a wide range of hyperparameters, aiming to minimize RMSE. The search space and selected parameters are summarized in Table 6. A total of 50 trials were executed, with the optimal configuration determined based on the lowest RMSE on the validation set. Similarly, RF was subjected to a grid search with cross-validation, evaluating multiple hyperparameter combinations to identify the best-performing model. The complete search space is detailed in Table 6. The best hyperparameters were selected based on the lowest RMSE using grid search 5-fold cross-validation.

Data availability

Data will be made available on request.

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